

MarineGuard AI — Investigation Report

SYNTHETIC DEMONSTRATION DATA — not real-world observations.

1. Incident information

Field	Value
Incident ID	MG-2026-DEMO-001
Title	Synthetic Arabian Sea oil-slick investigation
Location	Arabian Sea, west of Mumbai (synthetic scene)
Observation time	2026-03-14T06:30:00Z
Provenance	SYNTHETIC DEMONSTRATION DATA

2. Satellite observation

SAR scene associated with this incident. Detection mode: **U-Net ML Inference**. Trained U-Net weights were used for segmentation.

3. Spill detection

Field	Value
Mode	U-Net ML Inference
Confidence	90.9%
Status	Completed

4. Spill geometry

Field	Value
Area	69.28 km²
Length	41.08 km
Width	1.69 km
Perimeter	85.48 km
Centroid	19.10004, 71.62822
Shape	elongated

5. Probable origin

Field	Value
Latitude	19.11639
Longitude	71.55862
Estimated origin time	2026-03-14T02:00:00Z
Window	2026-03-14T00:30:00Z → 2026-03-14T03:30:00Z
Uncertainty radius	5.44 km

Probable origin is an analytical backward integration combining surface current vectors and 3% wind leeway with Coriolis deflection. It defines a probable search window, not an exact spill location.

6. Hindcast result

Backward trajectory points: 25. Model: backward_euler_drift_integration.

7. Forecast

Lead time	Latitude	Longitude
+0 h	19.10004	71.62822
+6 h	19.09201	71.73006
+12 h	19.09031	71.82323
+24 h	19.0869	72.00959
+48 h	19.08008	72.38228

Forward trajectory forecast incorporates predicted hydrodynamic surface current and 3% wind leeway. Uncertainty corridor expands over time due to oceanic turbulence and atmospheric variation.

8. AIS traffic analysis

Field	Value
Total vessels	7
Spatial candidates	4
Temporal candidates	4
Final candidates	3

9. Candidate vessel ranking

Rank	MMSI	Score	Priority
1	419000444	76.4	HIGH
2	419000111	73.8	MEDIUM
3	419000222	67.4	MEDIUM

10. Evidence

MV NORTHBOUND (419000444) — HIGH-PROBABILITY INVESTIGATION CANDIDATE (Potential Suspect Vessel)

- Spatial compatibility: 56.0% (Closest approach: 15.4 km from probable origin region).
- Temporal compatibility: 100.0% (Transmissions align with the origin time window).
- Trajectory compatibility: 70.6% (Heading delta 55.3° relative to drift axis).
- Wind compatibility: 65.0% (Navigational state consistent with atmospheric leeway dispersion).
- Current compatibility: 90.0% (Positioning consistent with hydrodynamic current drift).
- Operating speed: min 14.4 kn, mean 14.5 kn.

MT SYNTHETIC STAR (419000111) — MEDIUM-PROBABILITY INVESTIGATION CANDIDATE (Candidate Vessel)

- Spatial compatibility: 55.0% (Closest approach: 15.8 km from probable origin region).
- Temporal compatibility: 100.0% (Transmissions align with the origin time window).
- Trajectory compatibility: 44.0% (Heading delta 148.2° relative to drift axis).
- Wind compatibility: 85.0% (Navigational state consistent with atmospheric leeway dispersion).
- Current compatibility: 90.0% (Positioning consistent with hydrodynamic current drift).
- Operating speed: min 0.2 kn, mean 5.0 kn. (Loitering detected)

MV COASTAL PEARL (419000222) — MEDIUM-PROBABILITY INVESTIGATION CANDIDATE (Candidate Vessel)

- Spatial compatibility: 74.5% (Closest approach: 8.9 km from probable origin region).
- Temporal compatibility: 100.0% (Transmissions align with the origin time window).
- Trajectory compatibility: 17.5% (Heading delta 156.7° relative to drift axis).
- Wind compatibility: 65.0% (Navigational state consistent with atmospheric leeway dispersion).
- Current compatibility: 70.0% (Positioning consistent with hydrodynamic current drift).
- Operating speed: min 11.8 kn, mean 12.0 kn.

11. Confidence

Detection confidence 90.9%. Drift uncertainty radius 5.44 km. Forecast corridor width 23.6 km.

12. Data gaps

- Satellite detection may include false positives, especially in prototype/demo mode.
- AIS coverage may be gapped, delayed, or spoofed.
- Ocean wind/current fields are sparse demonstration vectors, not a circulation model.

13. Responsible-AI disclaimer

This system prioritises vessels for investigation and does not establish definitive legal responsibility. The highest-ranked vessel is a highest-priority candidate only. Results require human validation and must not be used as the sole basis for legal enforcement.